

Advanced (Habanero)

- Q1.** A 10-gallon solution of 20% alcohol is mixed with a 5-gallon solution of 50% alcohol. What is the concentration of the resulting mixture?
(A) 25% (D) 136
(B) 30% (E) 146
(C) 28%
(D) 32%
(E) 35%
- Q2.** A chemist mixes x liters of a 15% acid solution with $(15-x)$ liters of a 25% acid solution. The resulting mixture is 20% acid. What is x ?
(A) 5
(B) 7.5
(C) 10
(D) 12.5
(E) 15
- Q3.** A 500-ml solution is 30% sugar. How much more water should be added to make the solution 20% sugar?
(A) 100 ml
(B) 125 ml
(C) 200 ml
(D) 250 ml
(E) 333.33 ml
- Q4.** A coffee shop mixes Colombian and French roast coffee beans. Colombian beans cost 5 per pound, and French roast beans cost 8 per pound. If the shop wants a 20-pound mixture that is 30% French roast, what will be the total cost of the mixture?
(A) 106
(B) 116
(C) 126
- Q5.** A pharmacist mixes a 5% ointment with a 15% ointment. How many grams of each should be mixed to get 120 grams of 10% ointment?
(A) 60 grams of 5%, 60 grams of 15%
(B) 40 grams of 5%, 80 grams of 15%
(C) 80 grams of 5%, 40 grams of 15%
(D) 50 grams of 5%, 70 grams of 15%
(E) 70 grams of 5%, 50 grams of 15%
- Q6.** A 200-gallon tank is filled with a 10% saline solution. 50 gallons of the solution is removed and replaced with 50 gallons of a 30% saline solution. What is the concentration of the resulting mixture?
(A) 12%
(B) 14%
(C) 16%
(D) 18%
(E) 12.5%
- Q7.** A solution of 8 liters of 50% alcohol is mixed with x liters of 20% alcohol. The resulting solution is 40% alcohol. What is x ?
(A) 4
(B) 6
(C) 8
(D) 12
(E) 16
- Q8.** A mixture of two types of soil, one with 20% clay and the other with 50% clay, is used to fill a garden bed. If the resulting mixture has 30% clay, what is the ratio of the first type of soil to the second type of soil?
(A) 1 : 1
(B) 2 : 1
(C) 3 : 1
(D) 1 : 2
(E) 1 : 3

Open Ended Questions (FRQ)

- Q9.** A 10-gallon tank is filled with a 15% acid solution. 2 gallons of the solution is removed and replaced with 2 gallons of a 40% acid solution. This process is repeated 3 more times. What is the concentration of the resulting mixture?
- Q10.** A chemist has two types of mixtures: one with 20% acid and the other with 50% acid. If x liters of the first mixture and y liters of the second mixture are combined, the resulting mixture is 30% acid and has a total volume of 20 liters. If $x + y = 20$, find the value of x and y .

Chef's Tips

- Always read the question carefully before answering
- Show all your work for free response questions
- Check your calculations for multiple-choice questions